

Heat Transport in a Two-Layer Soil Profile

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Introduction

A one-dimensional transient heat transport simulation was performed using DRUTeS for a two-layer soil profile consisting of Clay (bottom) and Clay Loam (top) with a total depth of 1.05 m. A constant water content of $\theta = 0.36$ was assumed. The simulation period was 48 hours (172800 s), with a constant temperature at the lower boundary (12°C) and a time-varying heat flux applied at the surface.

Tasks

1. Compute thermal conductivity for both soil layers.

Thermal conductivity was calculated using:

$$\lambda(\theta) = b_1 + b_2\theta + b_3\theta^2$$

For Clay Loam:

$$\lambda = 0.35 + 1.45(0.36) - 0.55(0.36)^2 = 0.80072$$

For Clay:

$$\lambda = 0.38 + 1.40(0.36) - 0.50(0.36)^2 = 0.8192$$

So Thermal conductivity are;

- Clay Loam: $0.80072 \text{ W m}^{-1} \text{ K}^{-1}$
- Clay: $0.8192 \text{ W m}^{-1} \text{ K}^{-1}$

2. Compute volumetric heat capacity of the soil-water mixture.

$$C = (1 - \theta)\rho_s c_s + \theta\rho_w c_w$$

For Clay Loam:

$$C = 2.2536 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$$

For Clay:

$$C = 2.2408 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$$

So volumetric heat capacity are;

- Clay Loam: $2.2536 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$
- Clay: $2.2408 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$

3. Solve the transient heat equation for the 48-hour simulation.

The governing equation:

$$c \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left(\lambda \frac{\partial T}{\partial z} \right)$$

The simulation shows a transient temperature response over 48 hours. The surface layer reacts strongly to the imposed heat flux, while deeper layers remain more stable. Heat propagates downward gradually with clear attenuation and time delay.

4. Apply the surface heat flux boundary condition.

A time-varying Neumann boundary condition was applied at the surface. This resulted in alternating heating and cooling cycles. The surface shows strong flux variations, while the 10 cm depth shows reduced and delayed response, indicating conductive damping. The surface flux has higher amplitude compared to 10 cm depth, confirming that thermal forcing is strongest at the boundary and weakens downward.

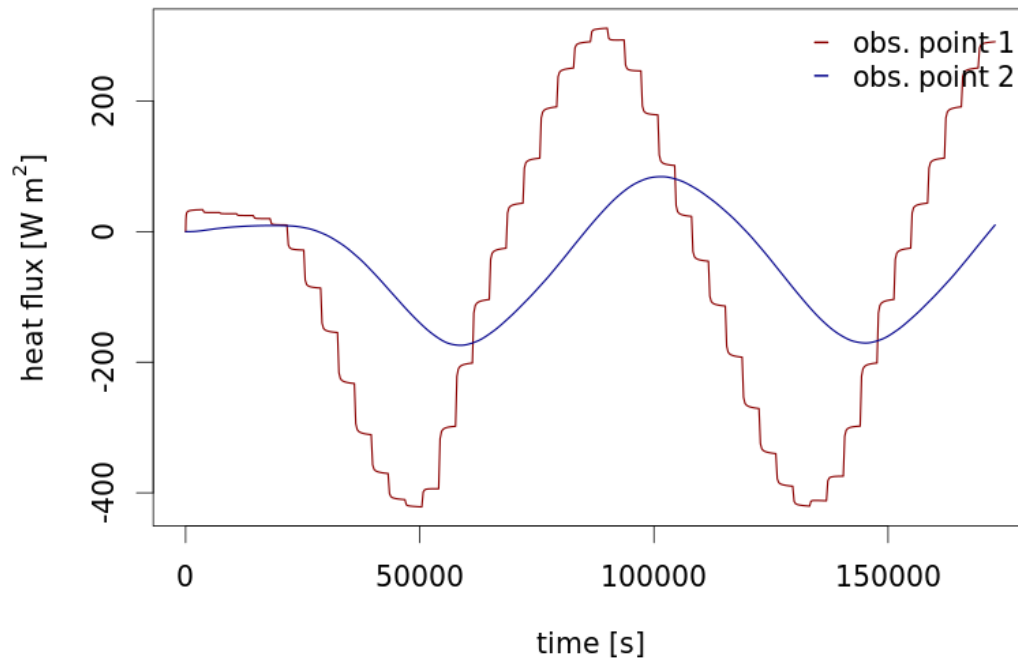


Figure 1. Heat flux at the soil surface and at 10 cm depth during the 48-hour simulation.

5. Assume constant water content during the simulation.

Water content was fixed at $\theta = 0.36$. Therefore, thermal properties remained constant during the simulation. The temperature evolution is controlled only by conduction and boundary conditions.

6. Compute the total energy inflow and outflow for surface and for the depth 10 cm.

From the surface heat flux:

- Total inflow:

$$2.5272 \times 10^7 \text{ J m}^{-2}$$

- Total outflow:

$$1.5948 \times 10^6 \text{ J m}^{-2}$$

The surface receives significantly more energy than it loses. At 10 cm depth, energy exchange is much smaller, showing that most thermal energy is absorbed or released near the surface. The cumulative curve at the surface is much higher than at 10 cm depth, indicating strong attenuation of heat transfer within the soil.

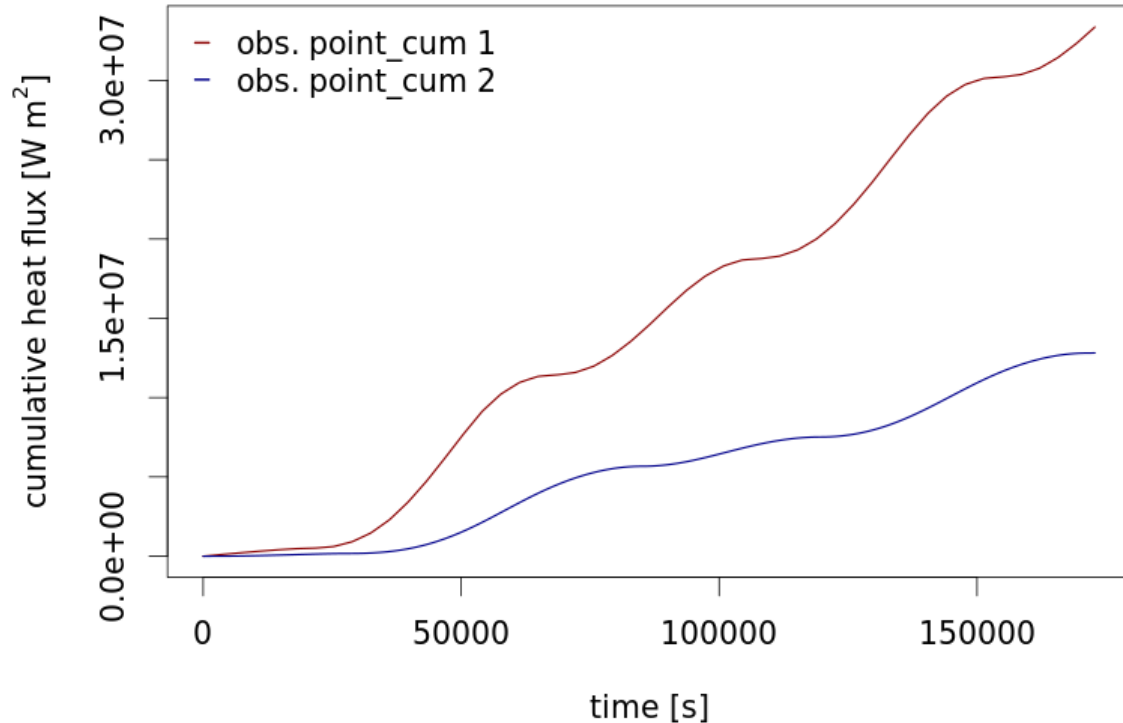


Figure 2. Cumulative heat transfer at the soil surface and at 10 cm depth.

7. Plot the temperature profile for simulation time 2, 10, 15, 48 hours.

- At 2 h: profile close to initial condition
- At 10 h: warming appears near surface
- At 15 h: strongest heating observed
- At 48 h: maximum surface temperature, deeper soil remains stable

Heat transfer is strongest near the surface and decreases with depth. In another words, temperature gradients become steep near the surface, while deeper layers show limited variation, confirming damping and delayed heat propagation.

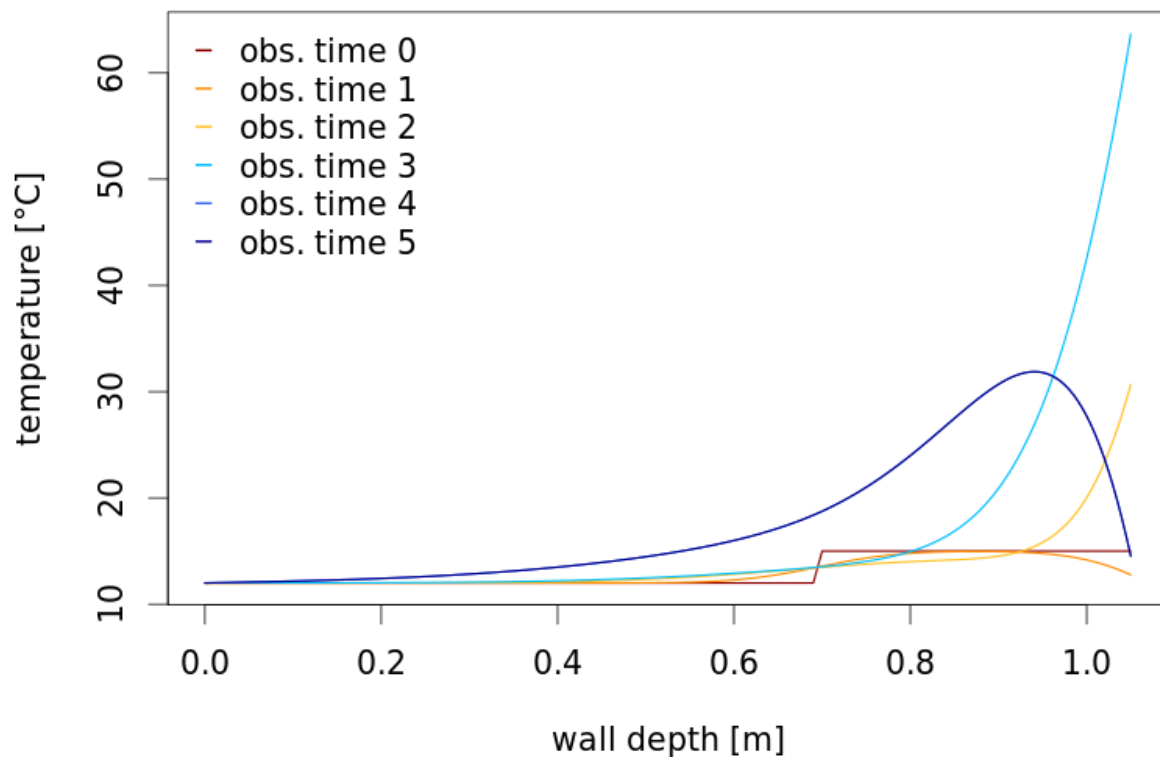


Figure 3. Temperature profiles at selected times during the 48-hour simulation.

Conclusion

The results demonstrate that transient heat transport is strongly controlled by surface forcing. Most energy exchange occurs near the surface, while deeper layers remain thermally buffered. Both flux and temperature outputs confirm progressive reduction of heat with depth.